

Experiment No. 7:

To determine the specific resistance of the material of a wire by a Meter bridge.

7.1 Objectives

To find resistance of a given wire using a meter bridge and hence determine the resistivity (specific resistance) of its material.

7.2 Prelab

Students must have a clear idea about the objectives and related theory of the experiment.

7.3 Outcomes

After completing this lab work, students will be able to answer the following questions:

- What is specific resistance?
- What factors depend on it?
- What is the principle of a Wheatstone bridge network?

7.4 Timing and Length of Investigation (Total 3 Hours)

- **Lab Preparation (15 minutes):**
 - For attendance and equipment distribution among the groups.
- **Lecture on Theory (40 minutes):**
 - The teacher will clarify the objective and theory of the experiment.
- **Lecture on Procedure 30 minutes):**
 - Demonstrator will make the students understand the procedure of the experiment.
- **Experimental Work (80 minutes):**
 - Design of experiment: 10 minutes
 - Data collection: 30 minutes
 - Data analysis, drawing graphs and completing the result part: 40 minutes.
- **Post Lab Discussion (15 minutes):**
 - The teacher will summarize the total lab work and have a discussion with the students related to the questions given in the outcomes part.
- **Report Submission:**
 - Students will prepare their lab report and submit it on the next class.

7.5 Theory

Meter bridge apparatus is also known as a slide wire bridge. It is fixed on the wooden block and consists of a long wire with a uniform cross-sectional area. It has two gaps formed using thick metal strips to make the Wheatstone's bridge.

In the arrangement as shown in Fig. 7.1, if X and R be the unknown and known resistance respectively and l be the distance of the null point measured from the left end A of the meter bridge, then by the principle of the Wheatstone's bridge network we get,

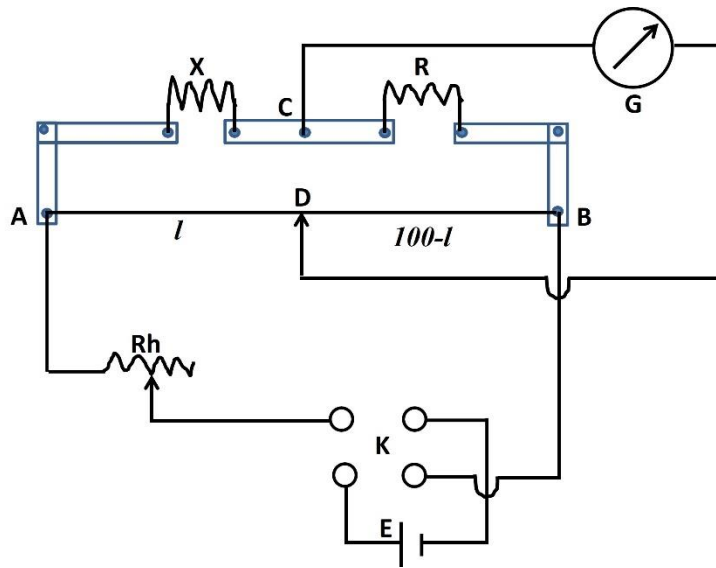
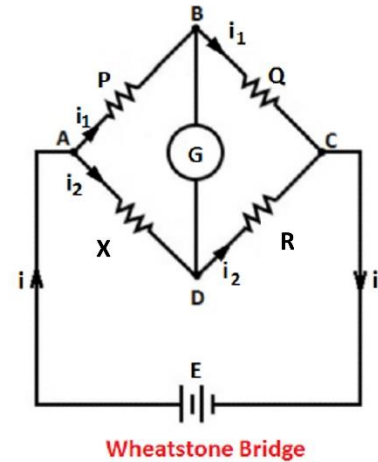


Fig.7.1: Circuit diagram for the experimental setup.



$$\frac{P}{Q} = \frac{X}{R} \text{ (Wheatstone bridge circuit)}$$

$$\frac{X}{R} = \frac{l}{100-l} \text{ (Meter bridge circuit)}$$

$$\text{Or } X = \frac{Rl}{100-l} \dots\dots\dots (1)$$

When the resistances X and R are interchanged, we get

$$\frac{R}{X} = \frac{l}{100-l}$$

$$\text{Or } X = \frac{R(100-l)}{l} \dots\dots\dots (2)$$

The mean of (1) and (2) will give the value of the unknown resistance.

Now if L be the length of the experimental wire, then

$$X = \frac{\rho L}{\pi r^2}$$

$$\text{Or } \rho = \frac{X\pi r^2}{L} \dots\dots\dots (3)$$

Where ρ is the specific resistance of the material of the given wire and r is the radius of the cross-section of the wire. Therefore, ρ can be determined after measuring the value of X, r and L.

7.6 Apparatus:

A meter bridge, A Leclanche's cell, A galvanometer (G), A resistance box (R), A jockey, Commutator (K), Rheostat (Rh), A specimen wire (X), A screw gauge, A meter scale, Connecting wires etc.

7.7 Procedure

1. Make connections as shown in the circuit diagram (Fig.7.1).
2. The wire whose resistance is to be determined should be connected in the left gap between A and C without any formation of loops.
3. The resistance box should be connected in the right gap between C and B.
4. Make all the other connections as shown in the circuit diagram.
5. The jockey should be first touched gently to the left end and then to the right end of the bridge.
6. The deflections in the galvanometer should be in opposite directions and if it is in one direction then the circuit connections are not correct. Move the sliding contact along the bridge wire until the galvanometer deflection is almost zero. Null point is being approached.
7. Let D be the null point where the jockey is touching the wire. Note the position of D to know the length of $AD = l$.
8. Reverse the current by changing the commutator plug and note the null point again. Take the mean of the two readings. From the relation of (1) calculate the value of X.
9. Now interchange the position of X and R with R in left gap. Find the the null point and reverse the current to find null point again. Calculate the value of X from relation (2).
10. Repeat the operations several times and calculate the mean value of X.
11. Carefully determine the length L and diameter of the wire by using a meter scale and a screw gauge respectively.

7.8 Observations:

A. Length of the wire $L = \dots\dots\dots \text{cm}$

B. Radius of the wire (r)

L.C of the screw gauge = $\dots\dots\dots \text{cm}$.

Observations	Main scale reading (cm)	Circular scale reading (cm)	Total reading (diameter) (cm)	Mean diameter (cm)	Mean radius (cm)

C. Readings for the balance point to calculate unknown resistance X.

Known resistance R (ohm)	Position of		Balance point for l			100- l	X (ohm)	Mean X (ohm)
	Unknown resistance X	Known resistance R	Direct	Reverse	Mean l			
	Left	Right						
	Right	Left						
	Left	Right						
	Right	Left						

D. Calculations:
Calculation for X

$$X = \frac{Rl}{100-l} \dots\dots\dots \text{(Left gap)}$$

$$X = \frac{R(100-l)}{l} \dots\dots\dots \text{(Right gap)}$$

$$\text{Calculate mean } X = \frac{X_1 + X_2 + X_3 + X_4}{4} = \dots\dots\dots$$

Calculation for specific resistance

Specific resistance of the material of the given wire,

$$\rho = \frac{X\pi r^2}{L} = \dots\dots\dots \text{ohm-cm}$$

Standard value of the specific resistance of the material of the given wire,

$$\rho_0 = \dots\dots\dots \text{ohm.cm}$$

$$\text{Percentage error} = \frac{\rho - \rho_0}{\rho_0} \times 100 \% = \dots\dots\dots$$

7.9 Results:

1. The value of unknown resistance $X = \dots\dots\dots$
2. The specific resistance of the material of the given wire = $\dots\dots\dots$
3. Percentage error = $\dots\dots\dots$

7.10 Discussions: